Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"
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SYSTEM QUALITY ASSURANCE DOCUMENT

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List of Acronyms

AD	Applicable Document
ASCAT	Advanced Scatterometer
BA	Burnt Areas
CDSE	Copernicus Data Space Ecosystem
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land OPerationS
CLMS	Copernicus Land Monitoring Service
DAG	Directed acyclic graphs
DB	Database
DMP	Dry Matter Productivity
ECV	Essential Climate Variable
EODC	Earth Observation Data Centre
ESA	European Space Agency
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
FAPAR	Fraction of photosynthetically Active Radiation absorbed by the vegetation
FCover	Fraction of vegetation cover
FTP	File transfer protocol
GCOS	Global Climate Observing System
GDMP	Gross dry matter productivity
GIT	free and open-source distributed version control system
GOES	Geostationary Operational Environmental Satellite
GPP	Gross Primary Productivity
GRD	Ground Range Detected
GRDH	Ground Range Detected High Resolution
GSA	GeoSphere Austria
GUI	Graphical User Interface
IDE	Integrated development environment
IPMA	Instituto Português do Mar e da Atmosfera
ISP	internet service provider
IW	Interferometric Wide swath mode
JIRA	A proprietary issue tracking product developed by Atlassian that allows bug tracking and agile project management
JRC	Joint Research Center
LAI	Leaf Area Index
LPV	Land Product Validation group
LST	Land Surface Temperature
LTO	Linear Tape-Open
METOP	Meteorological Operational satellite
MSG	Meteosat Second Generation

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NAS	Network-attached storage
NDVI	Normalized Difference Vegetation Index
NIR	Near infrared
NPP	Net Primary Productivity
NRT	Near real-Time
NTPD	Network Time Protocol Daemon
OData	Open Data Protocol
OLCI	Ocean and Land Colour Instrument
PAF	Processing and Archiving Facility
PID	Processing Information Datastore
PQAD	Product Quality Assurance Document
PROBA-V	Project for On-Board Autonomy – Vegetation instrument
PRTG	Paessler Router Traffic Grapher
QAR	Quality Assessment Report
QC	Quality Control
QFLAG	Quality Flag
RAM	Random Access Memory
REST	Representational state transfer
RSS	Really Simple Syndication
S1	Sentinel-1
S3	Simple Storage Service
SAN	Storage area network
SLSTR	Sea and Land Surface Temperature Radiometer
SQAD	System Quality Assurance Document
SSF	Surface State Flag
SSHD	Solid State Hybrid Drive
SSM	Surface Soil Moisture
SWI	Soil water Index
TB	Terabyte
TU	Technical University
TOC	Top Of Canopy
UPS	Uninterruptible Power Supply
UTC	Universal Time Coordinated
VITO	Flemish institute for technological research
VM	Virtual Machine

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GENERAL DEFINITIONS

Dry Matter Productivity (DMP) is an indicator designed to represent the growth rate of dry biomass: it is the amount (weight) of dry matter produced per surface unit and per time unit. The accumulation over time leads to the assessment of the biomass (in this context, phytomass), herein defined as above-ground mass per unit area of living plant material [GCOS-244]. The DMP is equivalent to the **Net Primary Productivity** (NPP) but has been adapted to agro-statistical use by transformation of the units into kilograms of dry matter produced per hectare and per day (kg DM ha^{-1} day^{-1}). Similar to the relation between NPP and DMP also the agronomic equivalent of the **Gross Primary Productivity** (GPP) can be defined as **Gross Dry Matter Productivity** (GDMP). The main difference between the NPP/DMP and GPP/GDMP is the inclusion of the autotrophic respiration.

Essential Climate Variable (ECV) is a variable or group of related variables that critically contribute to the characterization of Earth's climate state and forcing.

Evapotranspiration is the process by which water is transferred from the land to the atmosphere by evaporation from the soil and other surfaces and by transpiration from plants**.**

Fraction of Absorbed Photosynthetically Active Radiation (FAPAR) is defined as the fraction of Photosynthetically Active Radiation (PAR; solar radiation reaching the surface in the 400-700 nm spectral region) that is absorbed by a vegetation canopy [see GCOS-200].

Fraction of green Vegetation Cover (FCover) is defined as the fraction of ground surface covered by green vegetation as seen from the nadir direction (Baret et al. 2013).

Land Surface Phenology (LSP) describes vegetation changes observed from satellite remote sensing and constitutes therefore an aggregate signal from heterogeneous surface conditions that may not be representative of any single plant species response (Kimball 2014).

Land Surface Temperature (LST) is a measure of how hot or cold the uppermost surface of the Earth is. For ground-based, airborne and space-borne remote sensing instruments, it is the aggregated radiometric temperature of the ensemble of components within the sensor field of view [see GCOS-200].

Latent heat flux is the rate of energy transfer between the Earth surface and the atmosphere due to water changing phase in the process of evapotranspiration or condensation.

Leaf Area Index (LAI) is defined as the total one-sided area of all leaves in the canopy within a defined region, and is a non-dimensional quantity, although units of [m^2/m^2] are often quoted, as a reminder of its meaning [see GCOS-200].

Net Primary Production is the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day ($gC/m^2/day$).

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Normalized Difference Vegetation Index (NDVI) is the difference between maximum (in NIR) and minimum (round the Red) vegetation reflectance, normalized to the summation. The NDVI is an indicator of the greenness of vegetation.

Sensible heat flux is the rate of transfer of heat between the Earth surface and the atmosphere which results in temperature change of the atmosphere.

Surface Soil Moisture is defined as the relative water content of the top few centimetres soil, describing how wet or dry the soil is in its topmost layer, expressed in percent saturation. It is measured by satellite radar sensors and allows insights in local precipitation impacts and soil conditions.

Soil Water Index quantifies the moisture condition at various depths in the soil. It is mainly driven by the precipitation via the process of infiltration. Soil moisture is a very heterogeneous variable and varies on small scales with soil properties and drainage patterns. Satellite measurements integrate over relative large-scale areas, with the presence of vegetation adding complexity to the interpretation.

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) provides a series of bio-geophysical qualified products on the status and evolution of land surface at global scale. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

System quality constitutes all means to ensure that all systems (hardware, software, workflows, etc.) used to generate the products are working well and that timely action can be taken if a problem arises. In addition, it includes a basic check of the product quality, namely visual checks and simple statistics. These procedures are described in the System Quality Assurance Document (SQAD) while the methodologies applied for the exhaustive validation of the products are presented in the Product Quality Assurance Document (PQAD).

This SQAD details all the approaches followed to ensure the system quality for the three processing centres in charge of the production of the CLMS global “Vegetation and Energy” variables.

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1 BACKGROUND OF THE DOCUMENT

The System Quality Assurance Document focusses only on the methods put in place to monitor the quality of the system to produce the biophysical variables. The methods to perform the validation of the products and the annual scientific quality monitoring are described in the Product Quality Assurance Document (PQAD).

1.1 SCOPE AND OBJECTIVES

The SQAD presents the procedures applied by the consortium (CGLOPS-1) in charge of the production of the "Vegetation and Energy" variables to monitor the quality of the production before the products are disseminated.

This document does not include the procedures applied for the quality check of the "Water and Cryosphere" variables, which are under the responsibility of the CGLOPS-2 consortium.

Note as well that the procedures applied to ensure the quality of the product dissemination towards users are not considered in this document since the dissemination is outside the responsibility of the CGLOPS-1 consortium.

1.2 CONTENT OF THE DOCUMENT

Chapter 2 presents the CLMS "Vegetation and Energy" products.

Chapter 3 presents the overall quality monitoring strategy

Chapter 4 describes the operational quality monitoring applied in production centres.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

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1.3.2 Input documents

Document Id	Description
CGLOPS-1_PQAD	Product Quality Assurance Document of the Copernicus Global Land Operations Service “Vegetation & Energy”.

Available at https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document Id	Description
GCOS-200	The Global Observing System for Climate: Implementation Needs, WMO 2016
GCOS-244	The 2022 GCOS Implementation Plan

Available at https://library.wmo.int

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2 THE COPERNICUS LAND MONITORING SERVICE "VEGETATION & ENERGY" PRODUCTS

CGLOPS-1 consortium produces a set of global CLMS bio-geophysical products whose current technical characteristics are presented in Table 1 below. The NRT production is performed by three operation centres, each processing a category of satellite observations:

1. Polar orbiting optical sensors by VITO to generate the TOC reflectances, LAI, FAPAR, FCOVER, DMP/GDMP/NPP/GPP, NDVI, Burnt Areas (BA), Evapotranspiration and Heat fluxes
2. Polar orbiting microwave sensors by EODC, supported by GSA, to generate the soil moisture variables, i.e. the Surface Soil Moisture and the Soil Water Index
3. Geostationary sensors by IPMA to generate the Land Surface Temperature

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Table 1: CGLOPS-1 NRT portfolio on 21.09.2026

Product	Temporal coverage	Temporal resolution	Spatial coverage	Spatial resolution	Satellite Mission/Sensor
TOC reflectances	June 2018 - present	1 day	Global	300 m	Sentinel-3/OLCI & SLSTR
LAI – FAPAR – FCOVER – DMP – GDMP – GPP – NPP - NDVI	2014 - present	10 days	Global	300 m	PROBA-V + Sentinel-3/OLCI
Burnt Areas	Dec 2024 - present	1 day	Global	300 m	Sentinel-3/OLCI & SLSTR
	July 2018 - present	1 month	Global	300 m	Sentinel-3/OLCI & SLSTR
Land Surface Phenology	2014 - present	1 year	Global	300 m	PROBA-V + Sentinel-3/OLCI
Land Surface Temperature	2018 - present	1 hour, 10 days	Global	3 km	GOES-R+MSG+ IODC +Himawari
Surface Soil Moisture	2015 - present	1 day	Europe	1 km	Sentinel-1/CSAR
Soil Water Index	2015 - present	1 day	Europe	1 km	Metop / ASCAT + Sentinel-1 CSAR
	2007 – present	1 day, 10 days	Global	0.1°	Metop / ASCAT
Evapotranspiration	Nov 2025 - present	10 days	Global	300 m	Sentinel-3/OLCI
Heat Fluxes	Nov 2025 - present	1 day	Global	300 m	Sentinel-3/OLCI

3 OVERALL SYSTEM QUALITY ASSURANCE STRATEGY

The operational quality monitoring tasks check and ensure that occurrence of production happens according to timeliness and that product quality is as expected to allow the dissemination of products. The tasks are (semi-)automated such that a systematic check can be performed for each occurrence of the bio-geophysical variable.

The operational quality monitoring tasks are performed on the following levels:

1. Systematic hardware monitoring of the availability of the production system and its interfaces
2. Systematic software monitoring of the workflows to ensure timely product generation and delivery to the Copernicus Data Space Ecosystem (CDSE) for distribution towards users.
3. Routinely visual inspection to ensure that there are no major defects in the products
4. Routinely collect statistics (averages, median, quartiles, outliers) and respective error bars to identify physically unrealistic values.
5. Scientific quality monitoring verifies that the quality of the NRT bio-geophysical products stays stable along time compared to the results obtained during the exhaustive quality assessment

The results of the hardware and software monitoring tasks are gathered in reports in a (semi-) automatic manner, and hence can be consulted at any time.

The statistical tool is used both to analyse in NRT the behaviour of the product as it is generated through dynamical charts, with the possibility of visualizing past information, as well as to check the temporal consistency of the product.

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4 OPERATIONAL QUALITY MONITORING METHODOLOGIES

This chapter details the methods and procedures that are in place for the operational quality monitoring for each processing centre.

4.1 VITO PROCESSING CENTRE

4.1.1 System Monitoring

The technical quality of the system is daily checked by monitoring the hardware facilities through several tools and provides alerts for events like loss of connectivity, degraded performance, disk nearly full, etc. The actual monitoring is automated and hence carried out 365 days/year, 24 hours/day.

The monitoring tools keep logs to generate reports with at the following information for at least the last 6 months:

- Uptime of clusters or individual machines
- Average load of clusters
- Storage usage

The reports should prove a success rate of 95% or higher on the availability of the critical systems.

4.1.1.1 Hardware facilities

Processing and Archiving Facility (PAF)

VITO has an energy efficient Tier 2 data center. Redundancy is provided in terms of power for all critical components and cooling filled with computer racks, complemented with the latest networking technology, storage and power supply redundancy. The data center consists of 3 separate rooms, each blocked with fire resistant doors. The computer rooms have typically an area of 8 x 8m and are filled with racks, setup in a cold aisle containment. All computer rooms have power supply redundancy using a UPS system.

Storage infrastructure

VITO has set-up a hybrid scale-out Storage System that unifies its SAN and NAS infrastructure, this for the flexible storing and presenting of data towards the different processing chains and delivery services. The unique Netapp Storage Virtual Network technology guarantees a safe and performant

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presentation of the data towards all different network segments. The total capacity of this storage surpasses 2 Peta Byte.

For Long Term Data Archiving, VITO uses two remotely separated tape library systems supporting the latest LTO8 technology. These tape libraries have a (online) cloned archive available at all times. Data can easy been migrated from disk to tape and back. This system has centralized storage management included in the layer above the storage. The below figure (Figure 1) demonstrates the current setup of the storage network and different components used.

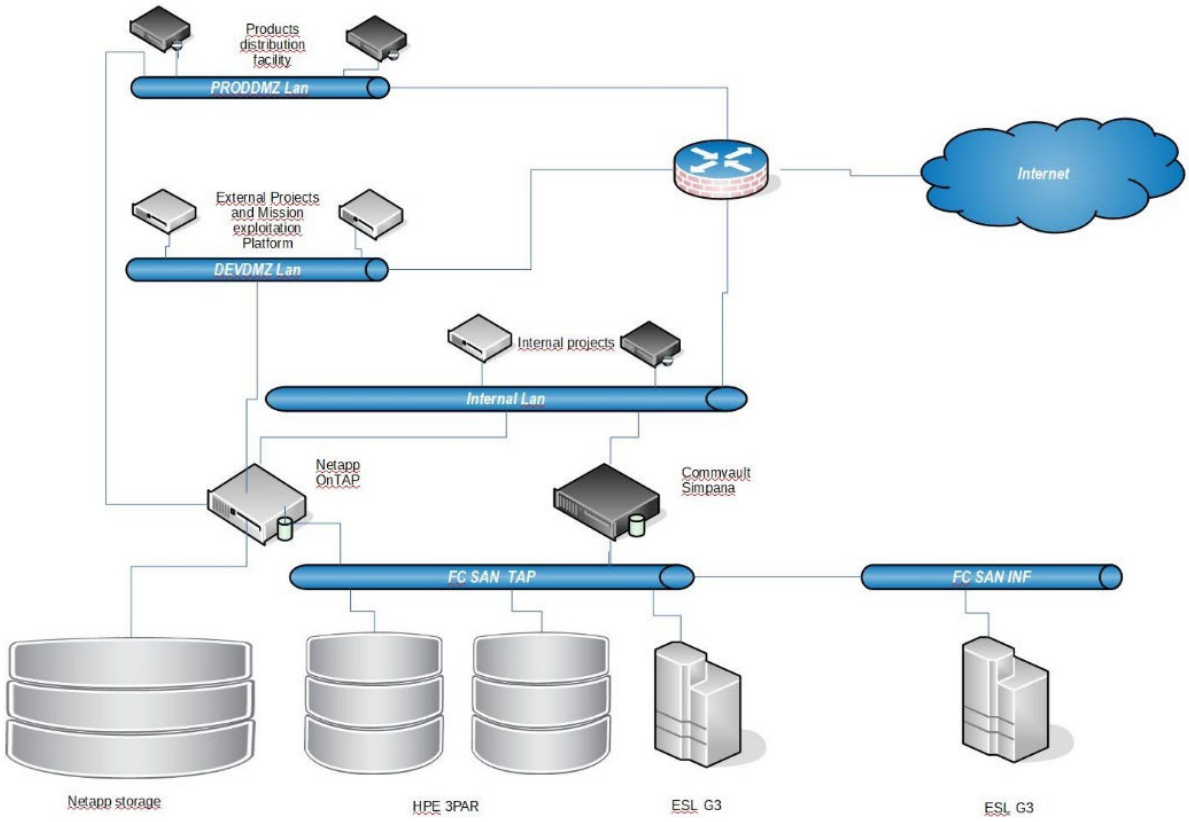


Figure 1: VITO Remote Sensing Storage infrastructure

Network infrastructure

VITO has separated network segments for development, test/validation and production. All production segments are separated from each other as well. VITO's internet connectivity has an upload/download capacity of 10Gbit/sec. VITO is connected via Belnet to GEANT/Internet2. To serve the end user community that is not connected to the GÉANT network, VITO has an internet link to the scientific and the commercial Internet of 10 Gbit/s, both being provided by the ISP Belnet.

The internal core network switching has a data rate of 40Gbit/sec towards a central redundant firewall and to 10Gbit/sec towards all crucial Top-Of-Rack switches in the data center.

High Performance processing cluster

This Hadoop cluster contains 7000+ cores and 27 TB of RAM which can dynamically scale in different queues depending on required processing requirements. Parallelization towards this cluster is managed by the Apache Spark framework. For Copernicus Global Land specific operations, a minimum of 33% of the cluster is reserved with a 100% availability when other processes are not consuming any of the other resources. The cluster ensures high availability through an automatic error rollover system. Tools such as Puppet, Docker and Terraform are used to deploy the full service into a cloud environment and to roll out changes in the configuration.

Due to the architecture and scalability of the Hadoop cluster, reprocessing can be invoked at any time and prioritized in a queue to avoid interference on near real time (NRT) processing. If NRT processing is idle, reprocessing queues will take over dynamically the resources of the NRT processing and vice versa, while always keeping priority for NRT processing. These queues and the amount of resources needed is configurable at any time.

4.1.1.2 System monitoring tools

Two major system monitoring levels are implemented:

1. PRTG networking tool for low-level infrastructure monitoring (servers/networking/storage),
2. Grafana (Figure 2) and Icinga to monitor the resources of the cluster and the services and applications running on the infrastructure.

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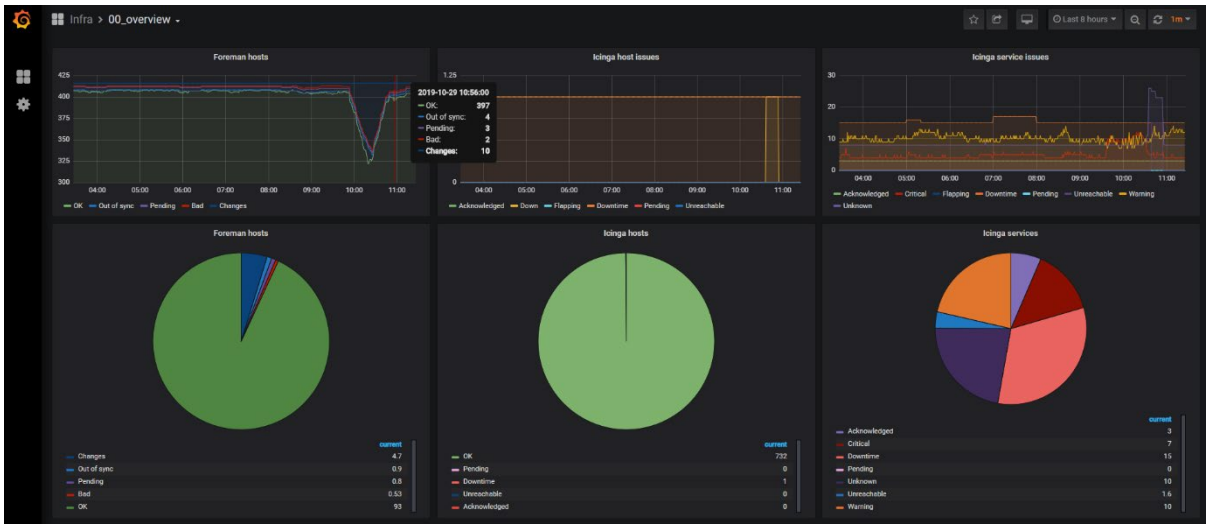


Figure 2: Grafana monitoring dashboard

4.1.2 Workflow Monitoring

A workflow or processing chain is best described as a sequence of individual processing steps. The individual steps are monitored through middleware software and information is logged for further reporting.

Each processing steps also logs the status and actions performed on the data it computes in the Processing Information Datastore (PID). The PID for product and processing steps is based on the open-source Elastic Search product. All processing steps log both their processing parameters (start time, end time, status, …) and the product status in Elastic Search. In addition, the processing step logs product status and processing kernel versions as metadata in the intermediate data files. By duplicating the data in Elastic Search and product metadata, a reliable, redundant decoupling of quality control and processing flow is achieved. As such, the processing flow can autonomously continue during high processing load while the PID is still processing the logged information.

The daily operation by experienced personnel contains a verification of the success for each processing step, the availability of output data/products, the uploading of products to the different distribution systems, the archiving of data/products and keeping the system in a clean mode, hence removal of non-used data/products.

Next to the operational monitoring by experienced personnel, the following logging information is kept for 6 months to generate operational statistics per 10-days (dekads) or per quarter year:

- % processing errors / processing jobs executed

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- Timeliness / variable, reporting the delay time from start periodicity (e. g. if daily products 0:00) and/or from the availability of the latest input data until the availability of the products per distribution channel (ingestion time)
- Number and volume of unique data products / variable
- Optional the number of pixels processed / variable
- Optional missing number (or volume) of input data products / variable
- Optional volume of input data used in Mbytes / variable
- Volume archived in Mbytes / variable

Incidents might trigger an update of the existing workflow software as a mitigation measure. Software updates are carefully planned and executed according to specific upgrade cycle procedure. This strongly reduces a disruption of the workflows and ensures product continuity.

Input data interruptions are handled in the same way as workflow interruptions. If a threshold is exceeded, this will be communicated further upstream towards Lot 3 to inform users of a potential delay.

4.1.2.1 Workflow monitoring tools

Apache Airflow is an open-source tool for orchestrating complex computational workflows and data processing pipelines. Workflows in Airflow are collections of tasks that have directional dependencies (see example in Figure 3). Specifically, Airflow uses directed acyclic graphs (DAG) to represent a workflow. Each node in the graph is a task, and edges define dependencies amongst tasks. The graph is enforced to be acyclic so that there are no circular dependencies that can cause infinite execution loops.

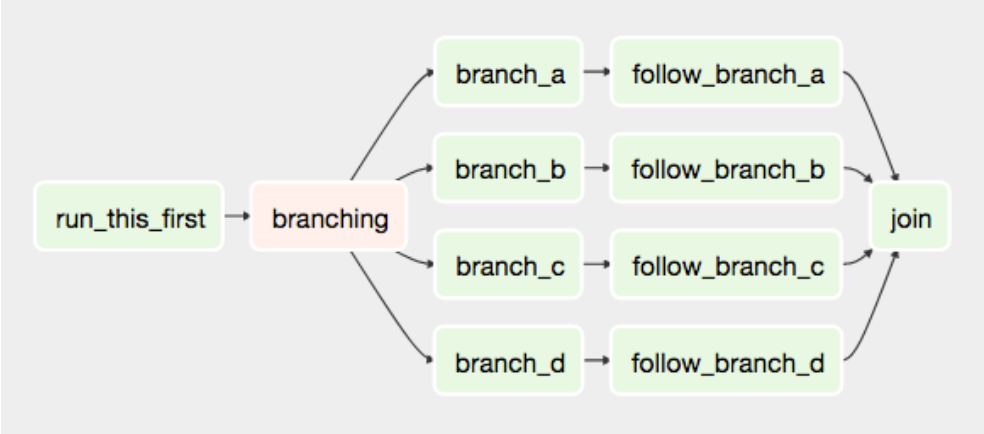


Figure 3: An example of Airflow DAG.

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Airflow makes it easy to monitor and troubleshoot workflow and data pipeline by offering components that support a full-featured Web User Interface (cf. https://airflow.apache.org/ui.html). This component allows a client-side user to view and edit the state of tasks in the metadata database. For instance, the workflow operator can see exactly how many tasks succeeded, failed, or are currently running at a glance (example in Figure 4).

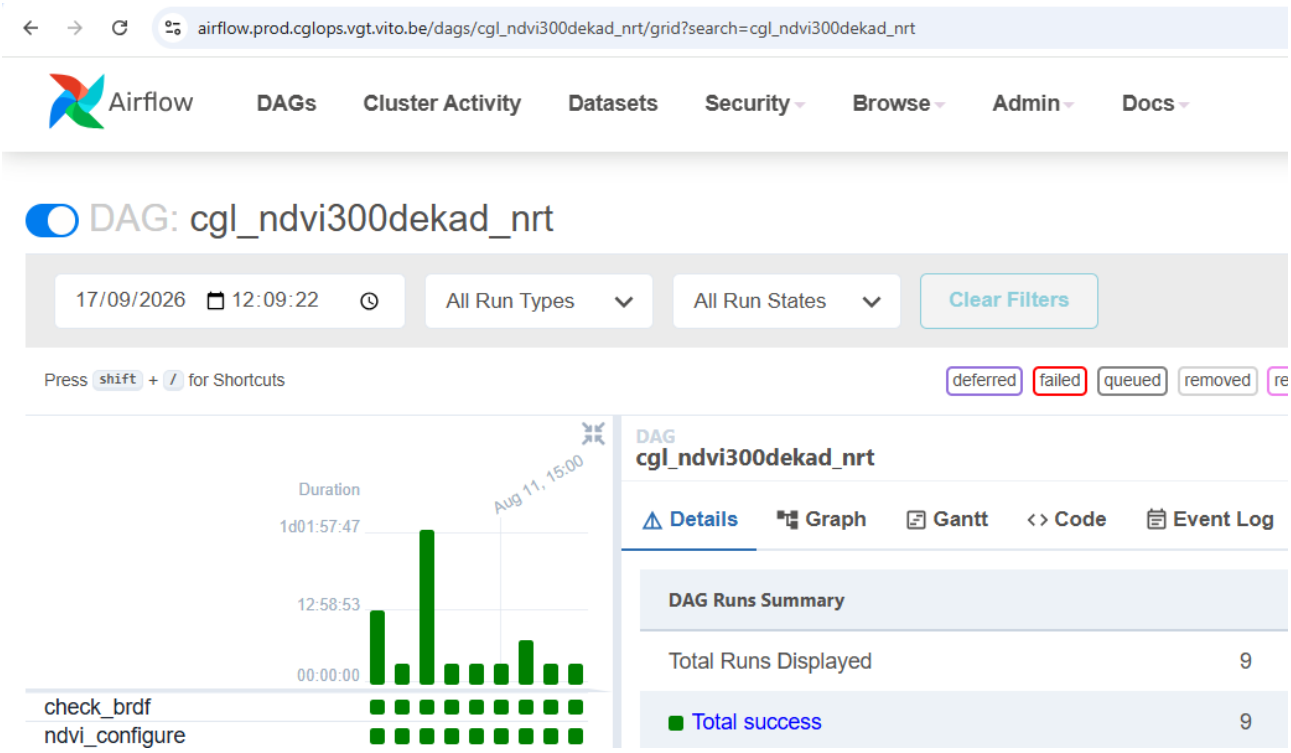


Figure 4: Airflow monitoring user interface.

If workflow steps that run on the Hadoop cluster fails, more detailed information of that job on the Hadoop system is obtained through the Hadoop Job Monitoring web interface. This interface details information on progress, the number of tasks running in parallel and the status of each task (Figure 5).

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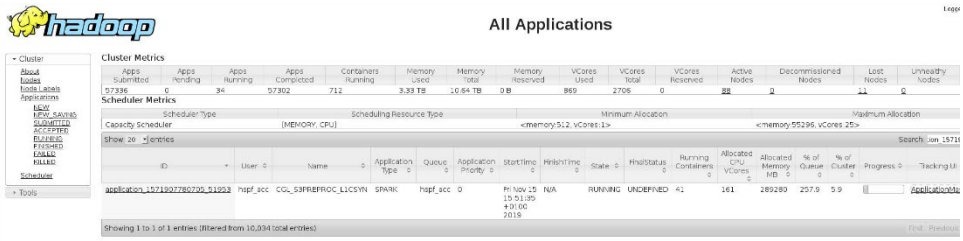


Figure 5: Job control monitoring

4.1.2.2 Software upgrade cycle procedure

Pre-configured Virtual Machines (VM) are offered to the algorithm developers and can run on the OpenStack cluster at VITO, providing the environment needed for them to work with the data and develop/deploy applications on the platform, i.e., containing Integrated Development Environments (IDE), a rich set of tools and access to the complete data archive.

Furthermore, users can customize this environment by downloading more data and tools. As such the algorithm developers have access to all input and intermediate datasets and can work in close cooperation with the workflow developers and hence enable a smooth transition between development and operations.

These VM's enable the workflow developers to develop and test their new or revised workflows in a local Hadoop Spark environment with limited resources, before performing an acceptance test on the cluster. Separate environmental settings are used to develop, to validate the functionality before acceptance in production, and to produce the product. Although they share the same hardware infrastructure, priority scheduling and the usage of Docker containers ensure that production is never compromised by development.

Deployment and testing are handled through a Jenkins server, which is coupled with the version control server (GIT) and the issue ticketing system (JIRA). This close coupling forces continuous deployment, ensuring that the code in the version control environment is exactly the software running in the workflows.

A versioning scheme follows the upgrade lifecycle and all phases in the lifecycle run in parallel in isolation, as shown in Figure 6.

Implementation

Here the actual code is written on a pre-configured VM. Although the code is version controlled (GIT) and supposed to be stable, it is certainly not reliable and very dynamic. In the development phase, automatic tests can be applied. These tests can be in the form of automatic unit-tests on the Jenkins server after a commit to GIT or integration test.

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Acceptance

The last part of a development step is creating a release candidate or patch that can be tested in the acceptance phase. The code is version controlled (GIT – local repository). The software and manuals are tested, verified and accepted by the operators. The produced output is used for the verification tests of the software and the algorithms. This verification is to be done by researchers in-house or researchers who delivered the algorithms. Important here is that the production that is done here, is meant for testing, not for distribution to customers. In this phase, only bug fixes are allowed.

Production

The software that is running in this phase is supposed to be stable and well tested. The software is version controlled (GIT Master repository). No software should be installed in live environments without passing through the acceptance phase. Operators are now using tested and validated software and procedures.

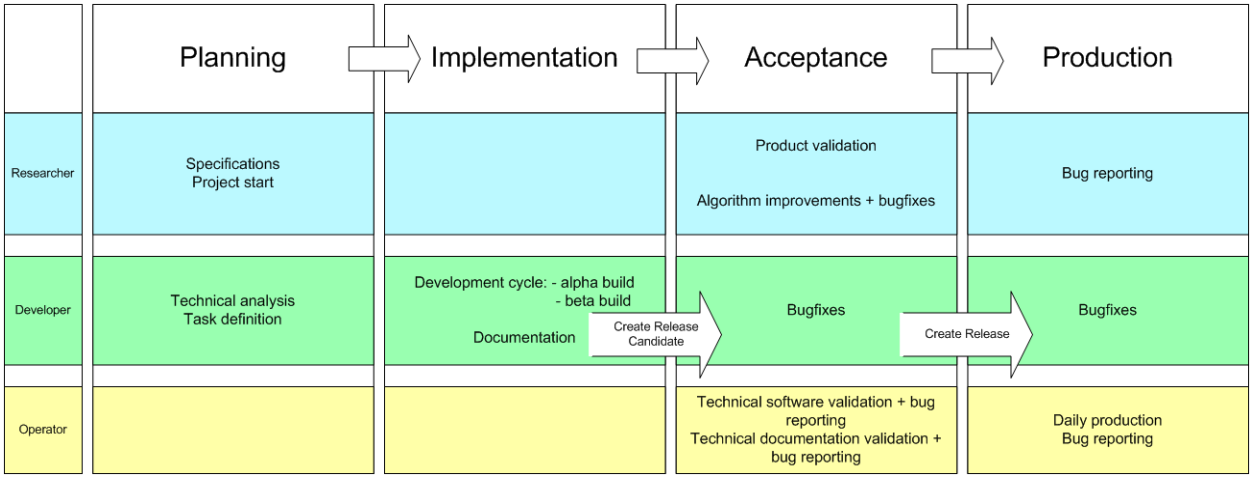


Figure 6: Upgrade lifecycle at VITO

The following tests are implemented at the various phases of the software lifecycle:

Unit testing

This section specifies the testing strategy and testing process followed for the unit testing of the processing modules. In this type of test, the module is viewed as a “black box”, which means that each module is unit tested in the following way:

- Test all nominal situations of the module with respect to its external interface (inputs and outputs) or internal variables.

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- Test robustness of each module with respect to erroneous inputs from its external interface.

The testing of the different processing modules will follow a bottom-up approach, allowing first developed module to be used for testing the latest ones. As such, most of the "Unit Testing" can be considered as "Integration Testing".

Integration testing

The integration testing activity consists of testing incremental assemblies of the different modules. Because of the bottom-up development strategy, the integration testing is similar to the Unit testing of a component.

In addition to this, the integration test also verifies dataflow between the different processing modules and tests if the output of one module can be used as input to the next module. Thus, the objective of the integration test is to prove that one module correctly uses the data provided by a previous module.

Verification testing

The following guidelines are used in the verification testing activity:

- All the verification tests shall be reproducible.
- Test cases are designed with the "black-box" technique, e. g. the test case cannot visualize the content of a variable in the code.
- Test data shall be re-used as much as possible.
- A set of "operational scenarios" are defined, representing all functionalities that are used in normal circumstances.

4.1.3 Visual Inspections

The result of critical processing steps may be checked through visual inspection to ensure that there are no major flaws (e. g. spatial or temporal discontinuities, registration errors, …). The visual inspection can be fully embedded in the processing chain in case of a low temporal frequency (>= 1 day) or outside the processing chain in case of a high temporal frequency (< 1 day). The visual inspection is done manually although we strive to perform as much as possible semi-(automatic) inspections to avoid delays in the processing. The result of a critical processing step is visualized by creating quicklooks that are inspected by an operator to prevent defects to be passed through resulting in faulty products (see e. g. Figure 7).

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If the temporal frequency of visual inspections is high (< 1 day), an inspection based on random sampling may be performed.

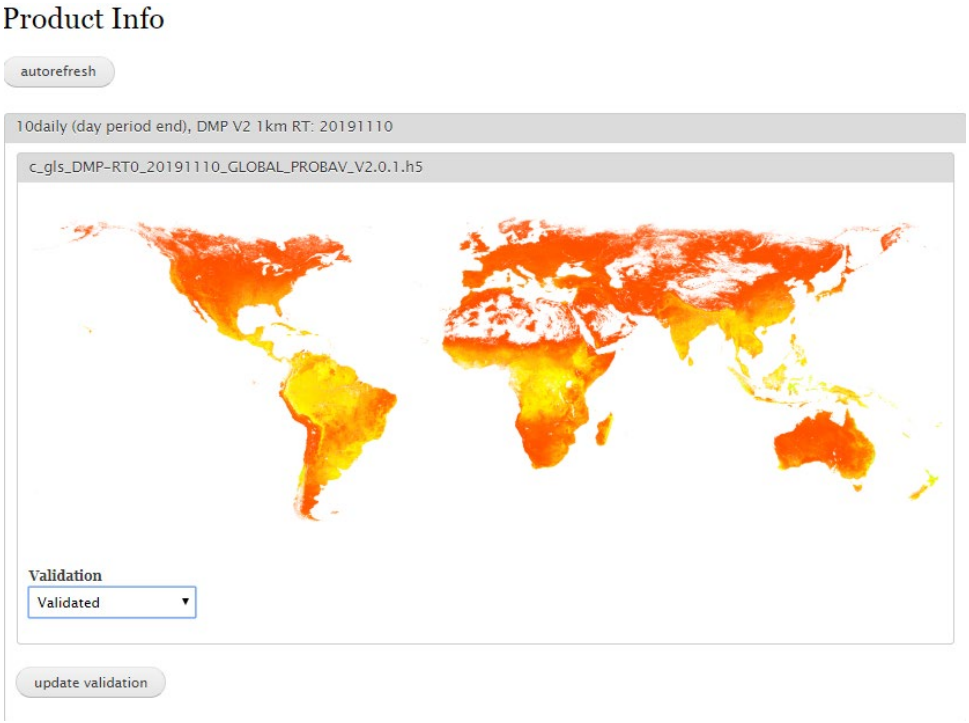


Figure 7: Examples of the visual inspection GUI

4.1.4 Statistical Inspections

The third type of monitoring is checking for unreliable product values by means of displaying statistical parameters of each generated product (i. e. for every timeslot and for each geographical region). The procedure is implemented operationally to analyse in NRT the behaviour of a product as it generates by means of visualizing the statistical parameters in dynamical charts. The average, median, first and third quartile, percentiles 5% and 95%, standard deviation, extreme values, rate of successfully processed pixels and rate of outliers with respect to land pixels are gathered on each product file.

In case of high temporal frequency products (< 1 day), the procedure that verifies the content of each product file for a specific region produces a 100 bins histogram together with some statistical parameters. At dekad (10-daily period) date, these daily histograms are used to calculate a single set of statistical parameters to represent the full dekad period.

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4.1.5 Operations, system maintenance and system failures

VITO data processing centre is an operational environment thus back-ups of personnel are inherent. This creates a maximum flexibility for the system as well as a motivating environment. This result in quality assured, flexible and motivated people, whereby fulfilling subtasks is possible. Four operators work 7 days a week to fulfil the operational services and manual product quality control. They are supported by software maintenance engineers during the workweek and on-call during the weekends. On the infrastructure side, there are four system administrators each having their own skill set but work together as team during the workweek and on-call during the weekends. All identified incidents are reported in a tracking system (JIRA) and are followed up through the predefined software life-cycle model.

On the operator side, weekly meetings are organized to discuss status and follow-up of pending issues. The operators work in a regime of 7 days/week for daily follow up of the workflows. The software maintenance engineers and system administrators work in an agile regime and have bi-weekly follow-up meeting with retrospectives and use a Kanban planning board (Figure 8) in JIRA for effective follow-up of the pending issues. In each of these status meeting, a representative of the other groups is present.

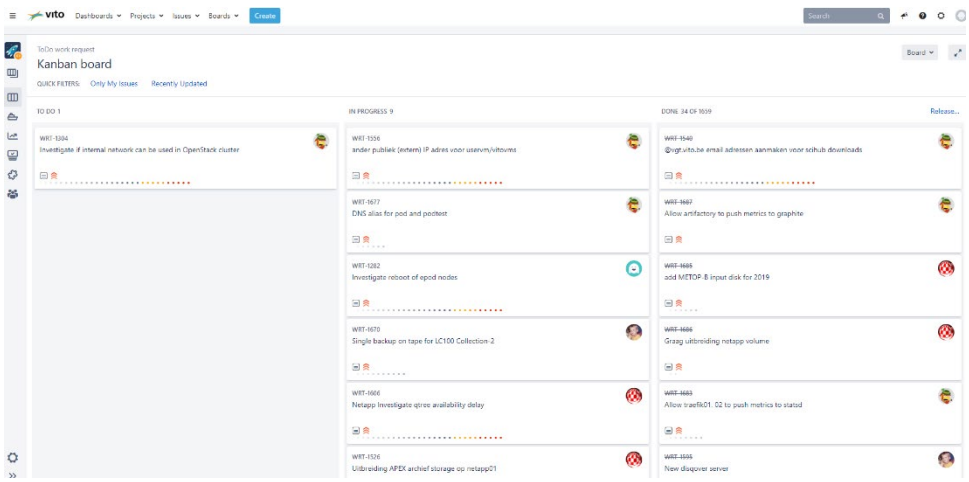


Figure 8: Example of Kanban follow-up board

Major system upgrades (e. g. network upgrades, electrical or cooling maintenance, etc.) are planned during holidays and in weekends. Minor system upgrades or planned updates with minimal service impact are usually planned before office hours (6:00-8:00) or during lunch breaks (12:00-13:00).

System maintenance and incidents follow the following procedures:

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- Processing and Archiving Facility, storage and network are maintained and supported by the System Administration Team.
- High Performance processing cluster is maintained and supported by the GeoMatics Team.
- The workflow specific system elements are handled according to the following procedure.
    - Operators will notice that a workflow is no longer active and will generate a JIRA issue describing the problem and linking to the corresponding workflow. The operators will then search the JIRA database if an identical incident has already been logged and resolved in the past. If this is the case and there is an operating procedure that the operators can perform, they will handle the issue. This is mainly the case in nominal system errors such as problems on the network, storage, etc.
    - VITO CGLOPS software engineers are automatically informed of a new incident. If the operators are capable of handling the incident, no further actions will be performed by the software engineer.
    - Otherwise, a software engineer is assigned to analyse and resolve the incident, following the software upgrade procedure.

Each component in a workflow can be restarted in order to be able to resume the interrupted workflow on the last successful workflow step.

4.2 EODC/GSA PROCESSING CENTRE

The EODC provides the operation centre for vegetation relevant products in collaboration with its subcontractor GSA. Using its cloud environment, EODC is responsible for the large volume pre-processing tasks associated with the acquisition of preparation of Sentinel-1 (S-1) data for the operational SSM and SWI production chains implemented at EODC.

To ensure robust and uninterrupted access to Sentinel-1 data, including GRD and complementary orbit data, the architecture relies on the Copernicus Data Space Ecosystem (CDSE) as the primary data access point, queried directly via its OData and S3 interfaces. In addition, EODC is part of ESA collaborative Ground Segment

CDSE is deployed as a cloud computing system which provides a Data Access element for end-user access to the Copernicus missions User Level Data. It is a critical component of the Copernicus Ground Segment architecture, ensuring reliable and secure data delivery.

CDSE differentiates between user categories, enforcing certain quotas and limitations concerning monthly transfer limits, requests per month, bandwidth limit per connection, or number of concurrent connections. However, EODC is part of the so-called Copernicus Service User category, where such limits are high enough to ensure stable and fast data access.

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At the core of the processing layer, EODC runs a dedicated instance of the open-source workflow orchestration and monitoring platform Apache Airflow. For CGLOPS-1, all required processing workflows, such as the downloading of Sentinel-1 scenes, Sentinel-1 pre-processing or the complete SSM and SWI production generation, are implemented as DAGs, defined in Python scripts and executed by Airflow.

4.2.1 System Monitoring

The system quality is daily checked by monitoring GSA hardware infrastructure using several tools providing alert notifications. In order to monitor the computers Remote Host Check a tool developed at GSA is used. The load values of the machine, the total number of processes, the presence of some key processes such as Cron, NTPD, Syslog, SSHD, etc., the memory and swap allocation, the allocation of the various memory areas and the network traffic are typically monitored. In individual cases, the regular arrival of data and the continuous processing of data is also checked.

GSA has developed the optimal network and satellite communication infrastructures in order to verify the nominal product generation and data dissemination. Furthermore, an infrastructure monitoring scheme has been implemented that monitors the status of operational chains that are essential for the continuity of operation activities.

Operational product generation is performed at GSA on Slurm clusters running under LINUX. This machine architecture was introduced recently and is expected to remain the basis for the near future. Furthermore, on the software side, the Copernicus production chains are in close accordance with algorithmic developments of TU Wien, using i. a. the GIT software version control approach and continuous integration testing suites.

GSA operates 24/7; during normal working hours, system administrators are available on-site to effectively work on urgent issues and requests that could hamper the nominal production.

EODC leverages many years of experience with a range of (open source) monitoring tools and reporting of operational systems. To ensure the required service availability EODC has implemented and relies upon a fully automated and redundant monitoring and alerting system.

The monitoring framework is built on state-of-the-art, open-source and well-established technologies:

- **Prometheus** collects hardware and system-level metrics (CPU load, memory usage, disk I/O, network throughput) from exporters on every relevant compute node, providing granular visibility into infrastructure health.
- **Airflow** orchestrates and monitors the SSM and SWI production workflow, giving visibility into task dependencies, execution times, and failure points.

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- **OpenTelemetry (OTel)** instruments the processing pipeline to collect traces and telemetry data, providing end-to-end observability across distributed tasks.
- **VictoriaLogs** provides centralized log aggregation and real-time log analysis, enabling rapid troubleshooting and anomaly detection.
- **Celery** enables distributed, asynchronous task execution across multiple nodes, ensuring scalability and resilience.
- **Grafana** provides interactive dashboards visualizing metrics, logs, and traces, giving operators and stakeholders real-time and historical insight into system health and availability.

To guarantee fast action, alerts are distributed through multiple redundant channels, including:

- **Slack** is a widely used team collaboration and messaging platform that supports real-time notifications, direct messaging, and integration with monitoring tools. Alerts from Prometheus and Grafana are pushed into dedicated Slack channels, ensuring immediate visibility for all team members. Slack also supports mobile apps, enabling notifications on smartphones for rapid response.
- **Microsoft Teams** provides similar functionality, offering persistent chat, group channels, and integration with alerting systems.

4.2.2 Workflow Monitoring

GSA, as a production center, ensures a reliable 24/7 operational product generation and distribution. Data dissemination, data services and their availability are automatically monitored in automatic procedures, in daily log files. Disruptions or arising disturbances create an automatic alert, where the 24-hour operation service at GSA can immediately react and eliminate/correct the error.

Currently, the NRT processing line for SWIv4 products (at 0.1° resolution) is operated at GSA (Figure 9). The chain is triggered daily at 16 UTC to process the 24-hour interval 12 UTC/yesterday-12 UTC/current day. The low production frequency permits to have a straightforward generation chain monitoring with providing daily product generation notification log files. After successful generation, the products are disseminated to an EODC server from where they are uploaded (by EODC) to the CDSE.

For monitoring the SWI 0.1° NRT production chain, an alert notification tool has been implemented that generates a daily log file; in case an error has been encountered, the file is sent by E-mail to relevant addresses to alert that the production chain requires attention, with the log file containing additional hints on the problem. A general estimate on the delay in operations in such a situation is not possible since it depends on the type of error.

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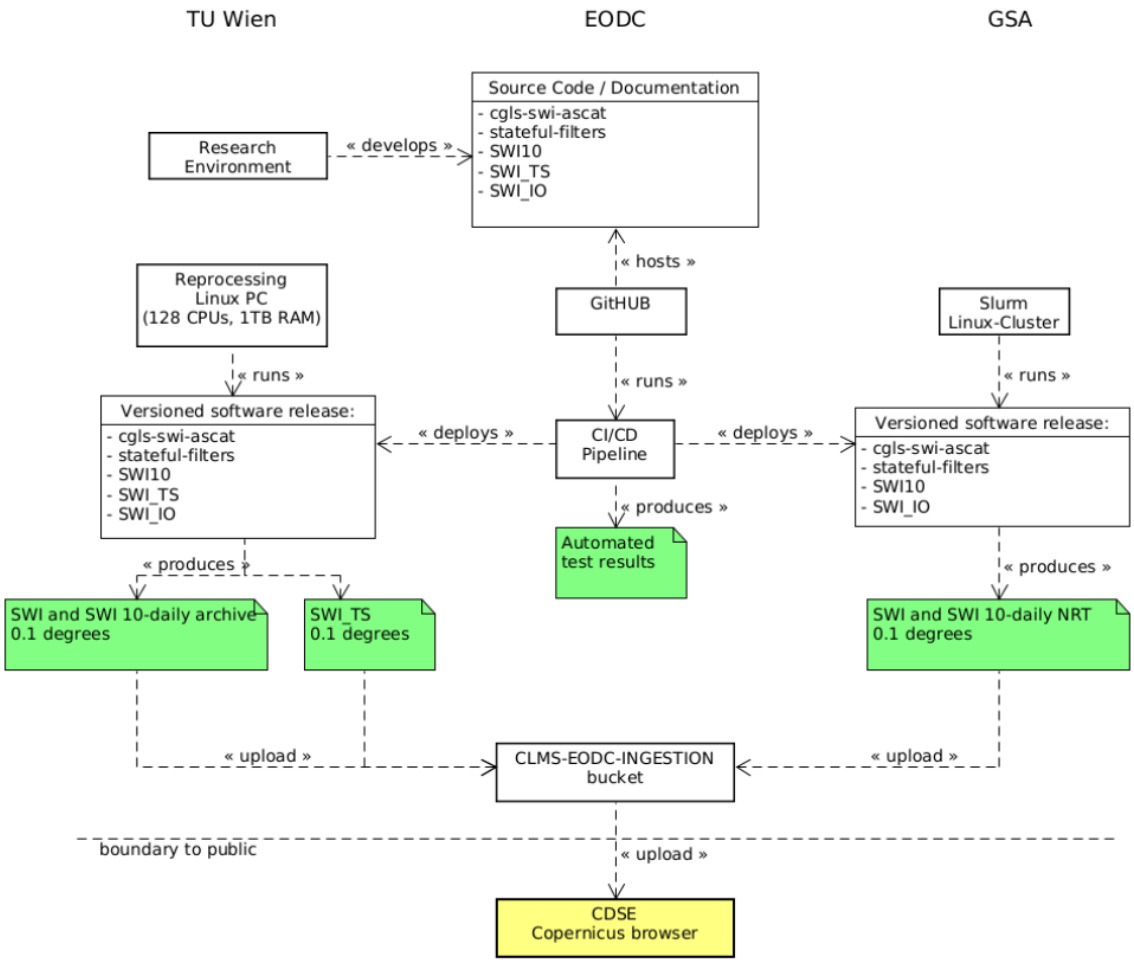


Figure 9: Overview of SWI 0.1° code and processing infrastructure across TU-Wien, EODC and GSA.

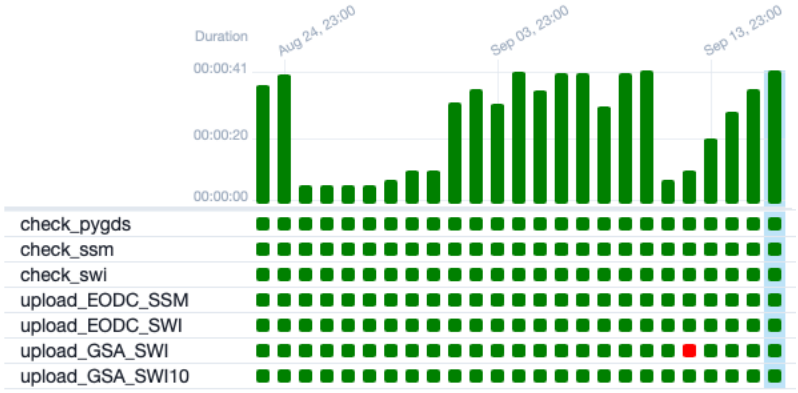


Figure 10: Monitoring interface in Apache Airflow

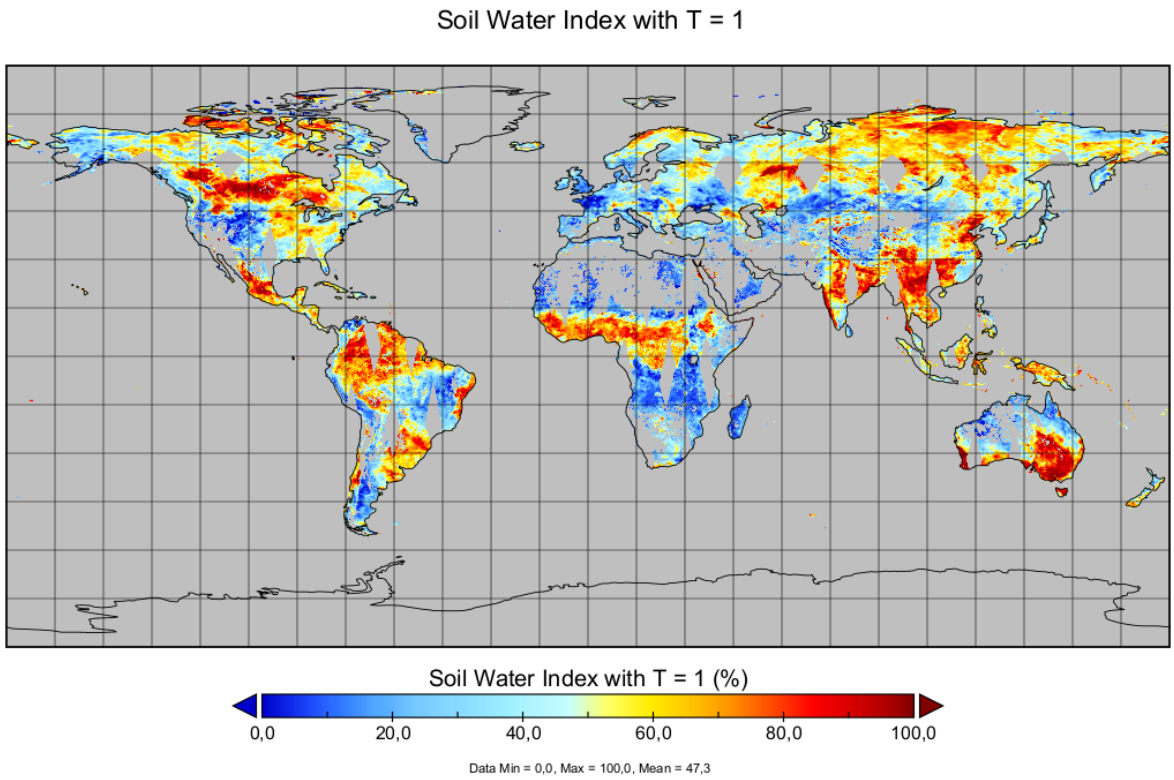
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To ensure the data pipeline runs correctly, a monitoring service is deployed within Apache Airflow to check the daily product generation. If the product is successfully created and exists in the target directory, the monitoring interface shows a green success state. Figure 10 shows an example from the currently (16/09/26) running NRT pipeline. Red states indicate an issue at the corresponding processing step.

Since the reason for production failure varies, for example missing input data, storage connection issues, or other problems, the EODC and, if required, GSA need to analyse the processing log file, input data, and related dependencies. After resolving the issue, a reprocessing activity can be carried out by manually rescheduling the processing task.

4.2.3 Visual Inspections

The SWI/SWI10 output files are in netCDF format, which means they can quickly be visualized for quality control by tools such as Panoply (Figure 11 provides example displays using this tool).



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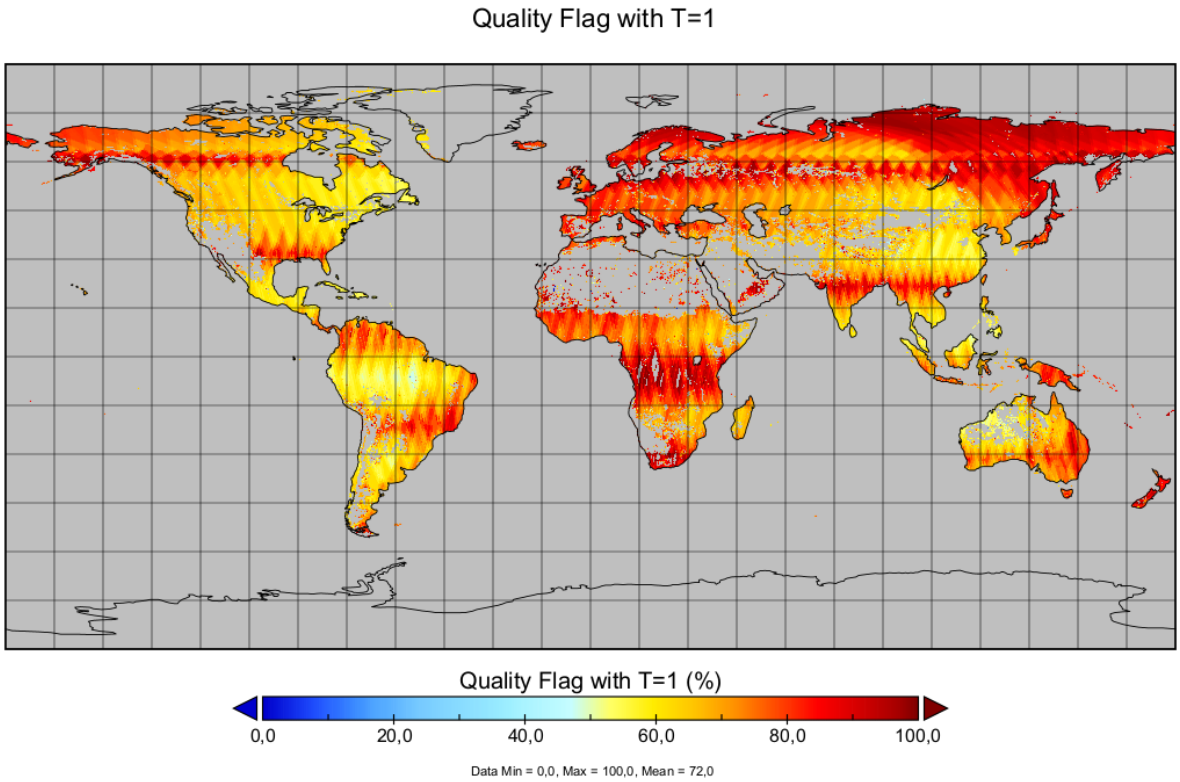


Figure 11: (Top) SWIv4.0.1 product visualization for 1 July 2026; this product is generated every day by the GSA operational centre and provides global daily information about moisture conditions in different soil depths. (Below) Quality flag for the SWI10 product that was produced on 30 June 2026, averaging the daily SWI product over the 10 days period 21/6/2026-30/6/2026.

4.2.4 Statistical Inspections

A quality control (QC) scheme is developed checking the integrity of the generated SWIV4 products, inspecting each SWI product values are within the expected nominal range. The quality control procedure detects emerging problems during data production that could affect the consistency of the SWIV4. Each operational SWI product inherently contains:

- 8 SWI and quality flag (QFLAG) values for various soil depths (characteristic T-values)
- Surface State Flag (SSF) that indicates frozen/unfrozen/melting state of the surface depending on the temperature. SSF is used to identify SSM values under non-frozen conditions to be used for SWI calculation.

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GSA maintains an internal graphical monitoring tool (snapshot in Figure 12), where any suspicious observation during the generation process alters the colour of the bar of the affected day from green to something else.

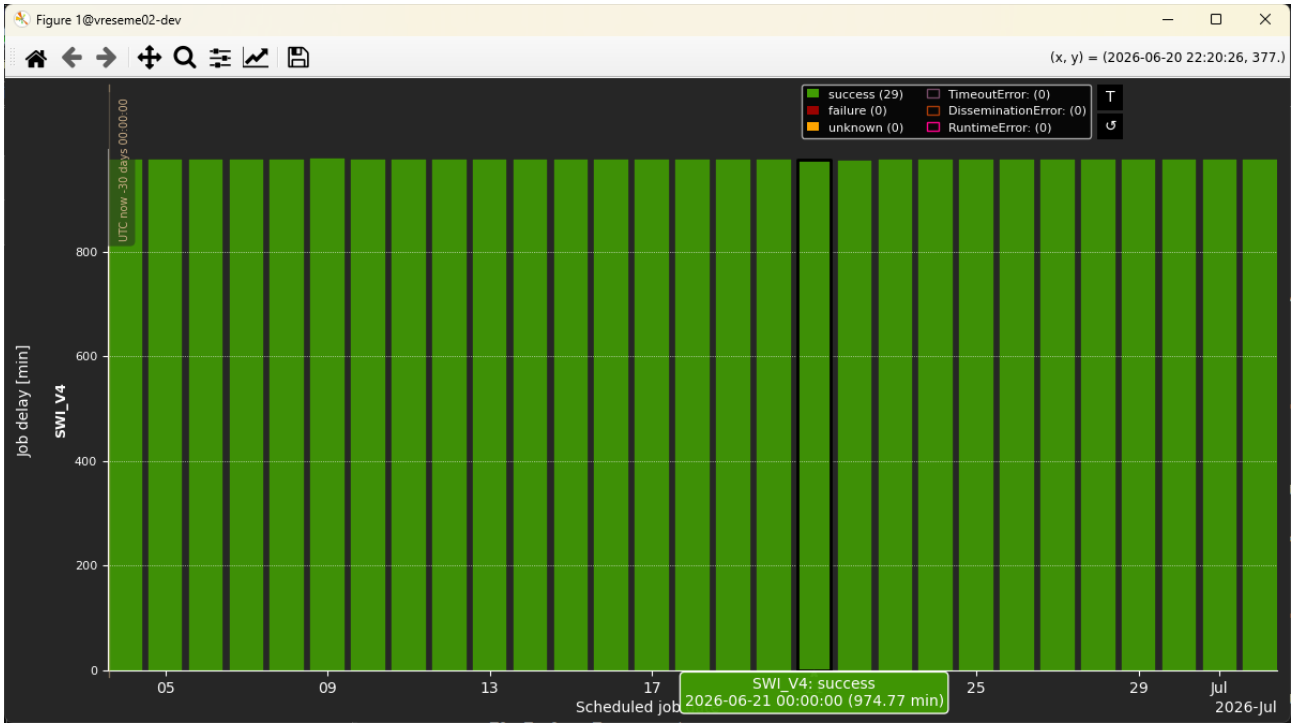


Figure 12: **GSA tool for quick monitoring of the recent behaviour of the local SWI production chain (Image confirms that there were no problems during June and there is no need for further analysis / troubleshooting).**

Checking and monitoring of the product timeliness is based on the tracking of metadata of the data products generated within the product generation system as well as on metadata automatically generated by the utilised processing platform Apache Airflow.

At EODC, after every reporting period (e.g. every quarter) a dedicated Airflow DAG produces statistics and visualizations of the respective timeliness. The processing platform Apache Airflow itself records metadata about the various task durations encompassed in the product generation workflow.

4.3 IPMA PROCESSING CENTRE

Like many other operational processing systems, IPMA's system is complex due in part to all the functionalities required and their dependencies. In order to boost system performance and availability, it is crucial to have an effective monitoring system that addresses all the essential aspects ranging from the hardware to the generated products. Doing so allows establishing benchmarks against which

is possible to measure performance metrics, determining the effectiveness of modifications introduced during the operational phase or even identifying trends.

The following monitoring steps are in place for the IPMA's system:

- Hardware monitoring, to monitor networking, disk space, disk i/o, load of the hosting machines and storage units;
- Input data reception monitoring, to monitor the reception of raw satellite and auxiliary data;
- Processing chain monitoring, to monitor all the processing and dissemination tasks status as well as to consult tasks dependencies and triggers;
- System monitoring, to assess the compliance of all the processing and dissemination activities with the timeliness requirements;
- Product quality monitoring, to identify strange patterns in the products generated or even product degradation with time.

All the monitoring tools assimilate system information and present it to the operators in a user-friendly way through intuitive graphs and/or numerical values. Moreover, some of statistics collected by the monitoring tools are also used to report on the system performance every quarter for service evaluation.

4.3.1 Hardware Monitoring

The Hardware monitoring relies on a back-end workflow based on *Graphite*, which is an open-source tool that stores numeric time-series of data in a *mySQL* database fed with data from each monitored machine via *collectd*. It also renders graphs on demand. However, to enhance the visualizations a graphite *frontend* (called *Tessera*) is configured. The Figure 13 provides an example of a *Tessera* dashboard analysed by the operations team during office hours.

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Figure 13: Tessera Dashboard for HW Monitoring.

4.3.2 Input Data Reception Monitoring

The arrival of mandatory input files events is monitored by *inotify*. This information is inserted via REST in a *sqlite* database rendered with a Django based application to display the availability of the input data in a calendar with an entry for each timeslot coloured in green, if all the expected files have arrived (even if delayed) or in red, if at least one file is missing. The number of arrived files is also displayed. Figure 14 presents as an example a 2 days view for GOES-R file reception, including the timeliness and files count, which is regularly analysed by the operators during office hours and as a first step on the investigation of failure detected at the System Monitoring (see Section 4.3.4).

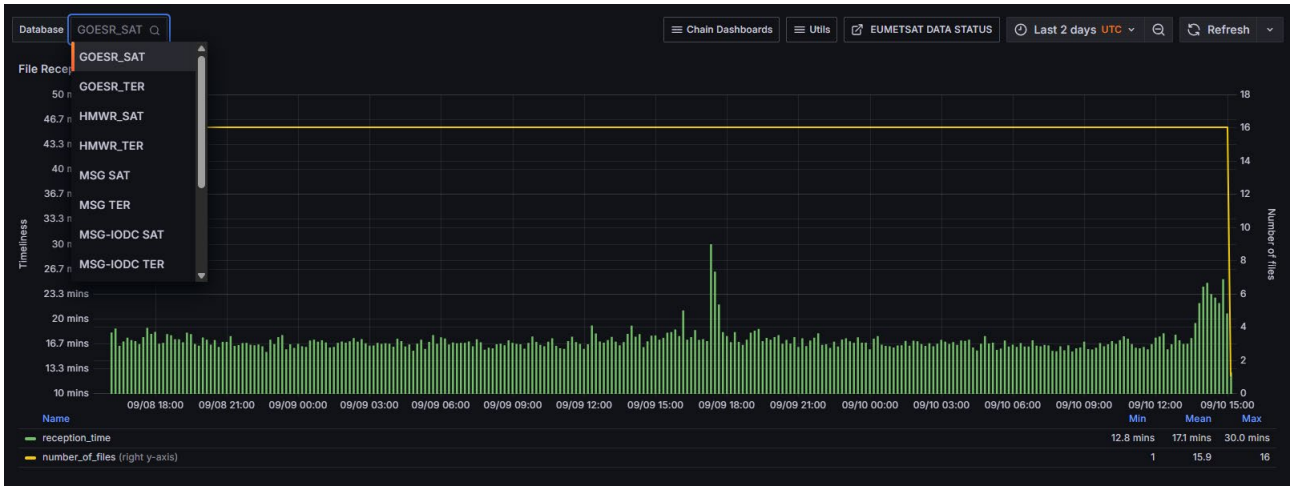


Figure 14: Screenshot of GOES-R raw data monitoring.

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4.3.3 Processing Chain Monitoring

The orchestration of the different tasks that composes the production chain relies on the *ecflow* which is provided with a GUI (*ecflowview*) allowing to monitor the status of each task and also to consult their dependencies and triggers.

Each task of the processing chain corresponds to a coloured box depending on the processing status. The Figure 15 presents a screenshot of the LST10 processing chain displayed in the ecflowview.

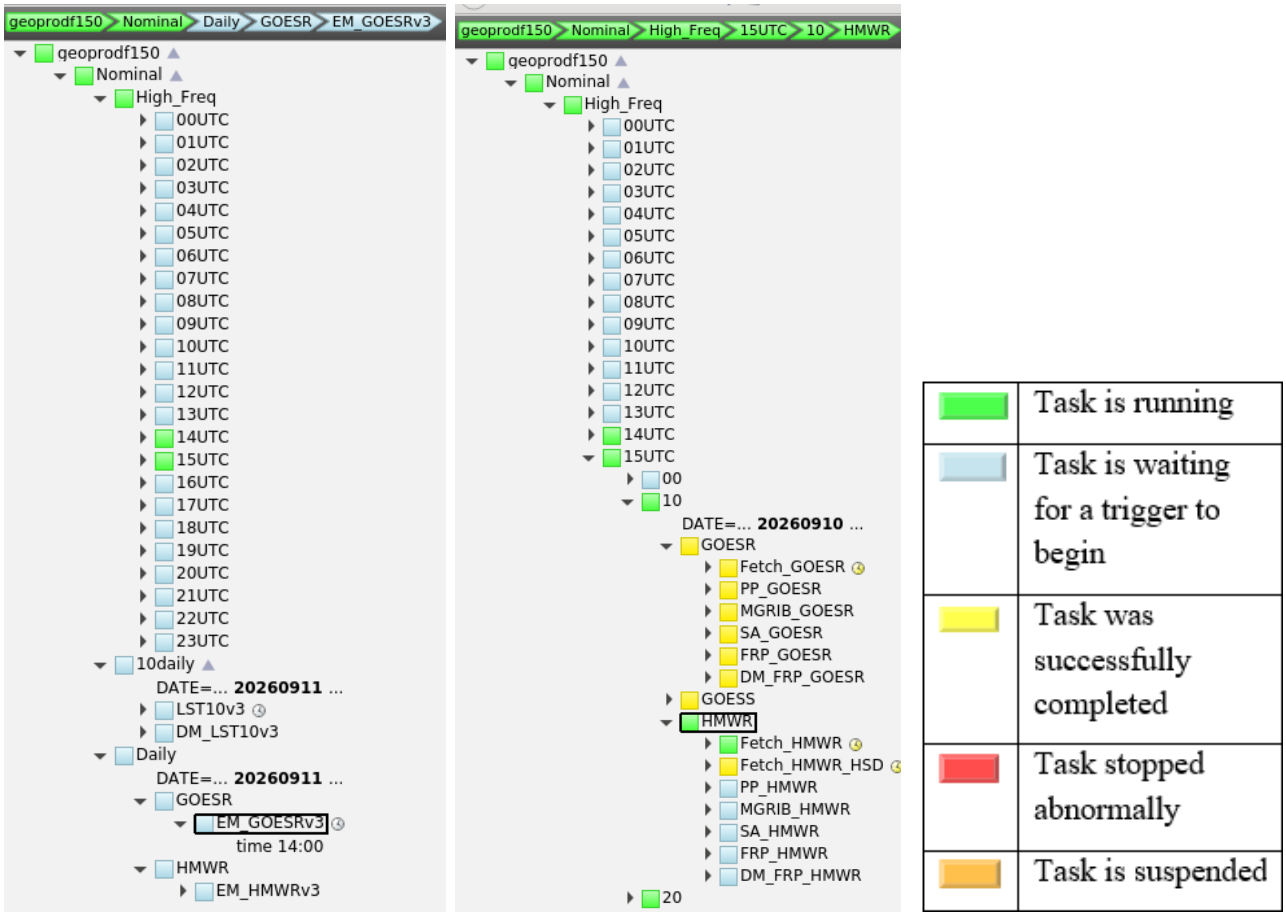


Figure 15: Screenshots of ecflowview showing the CGLOPS chain while executing the tasks for 2026/09/10 15:10 UTC. At right, ecFlow color code for processing status of individual tasks within a processing chain.

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4.3.4 System Monitoring

The real-time system monitoring is supported by customized web-based timeline that displays the status of both production and dissemination verifying if the respective timeliness is within the product requirements.

The timeline presents, for each timeslot and product, a coloured pie icon split into 4 quadrants each one indicating the status of the process and tasks distribution as described in the Table 2 and illustrated in the Figure 16. The timeline also presents a line with the current UTC time to help the operators to verifying which products should (or should not yet) be processed. The Figure 17 shows a print screen of the timeline available to the operators.

The main chart is composed by 2 graphs to display the set of products sharing the same time frequency:

- TOP Graph: for high frequency products (3-min, 15-min, 30-min and hourly);
- BOTTOM Graph: for daily products.

A default view showing the latest 12 hours is presented in the first access and after a page refresh. However, the most recent 2 days can be displayed by typing the “show all” icon at the top right side of the view.

The operators regularly verify if there are icons or pie slices with red or yellow in the timeline behind the current UTC time (presented as a yellow vertical line).

Table 2: Screenshot of the Monitoring Tool for CGLOPS Systems.

Pie slice	Description	Possible values/colours
First Quadrant	Status of **PRODUCTION**	Red – At least one product file was not produced or not inserted in the production DB Yellow – Product generated out of the defined timeliness Green – All the product files were successfully produced within the timeliness and inserted in the production DB Gray – Product not to be processed for this timeslot Blue – Product scheduled to be processed
Second Quadrant	Status of **EUMETCAST** dissemination	Not applicable

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Third Quadrant	Status of dissemination to the **FTP SERVER**	Red – At least one product file was not sent to the FTP Server Yellow – Product sent out of the defined timeliness Green – All the product files were successfully sent within the timeliness Gray – Product not to be disseminated Blue – Product scheduled to be processed
Fourth Quadrant	Status of reception of product back to IPMA facilities via EUMETCast (called **END2END** dissemination)	Not applicable

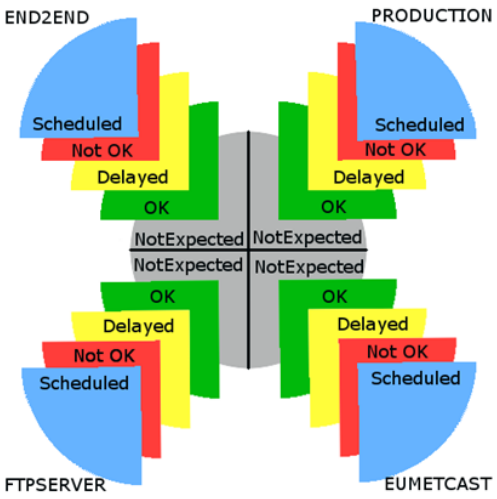


Figure 16: Legend of pie icons of timeline.

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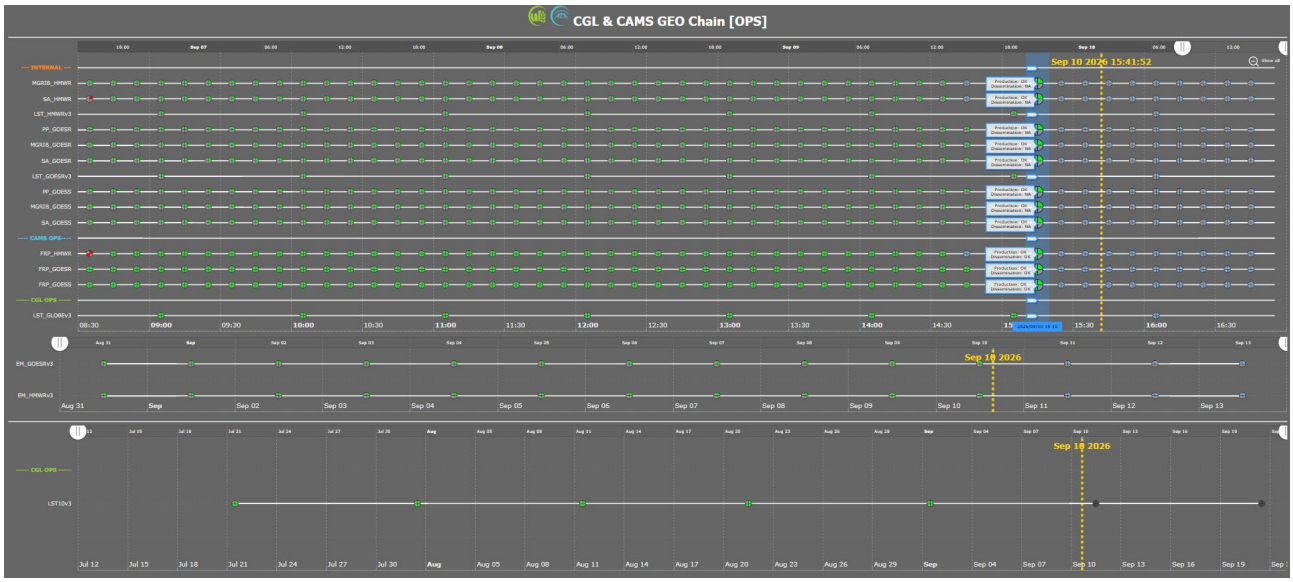


Figure 17: Screenshot of the System Monitoring Tool.

4.3.5 Product Monitoring – Statistical Inspections

On top of the system monitoring, also the products' quality is monitored by means of displaying statistical parameters of each generated product over each geographical region: North America, South America, Europe, Africa, Asia and Oceania.

For each generated product, an operational procedure is implemented to retrieve statistical parameters, such as the average, median, first and third quartile, percentiles 5% and 95%, standard deviation, extreme values, rate of successfully processed pixels and rate of outliers with respect to land pixels. Those statistical parameters are analysed on a weekly basis by the operators to check the temporal evolution of the spatial statistics and detect potential deviations or product degradation with time.

The Figure 18 shows an example of the dashboard visualized by the operators with the temporal evolution of the statistical parameters over a year using 10-day averages. The data can also be analysed over different time periods or using the original frequency.

The visual inspection of high-frequency products is performed when the operator detects a strange behaviour on at least one of the statistical parameters.

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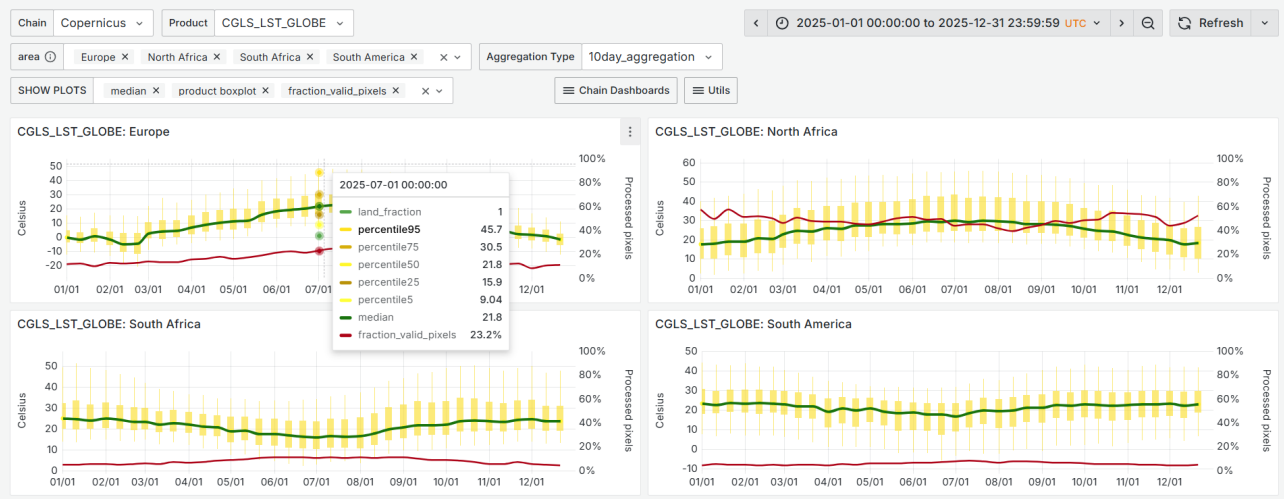


Figure 18: Screenshot of product quality monitoring showing the temporal statistics for hourly LST over a 1-year period. The time series represents the LST statistical distribution over 10-day aggregation periods, featured as yellow boxplots (the box shows the interquartile range and whiskers extend from the 5th to the 95th percentiles) and a green median line for each area. The red line shows the average fraction of processed pixels for each region.

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4.4 SCIENTIFIC QUALITY MONITORING

The objective of the scientific quality monitoring is to verify whether the recent products maintain the same quality as assessed in the exhaustive quality assessment described in the PQAD. The pillars of the scientific quality monitoring are:

1. **NRT information on product quality**. We compare the recent product with a reference data set derived from the product themselves. This reference data set can be either a reference year or a long/short term average of the product. Important is that the reference data set is the same data that was used in the exhaustive quality assessment. This allows to assess a number of quantitative metrics when a new product is generated. The metrics verify the completeness and the bias of the product with the reference data set for the same period (e.g. dekad).
2. **Easy to interpret and access**. The quantitative metrics feed into a dashboard on the website. This dashboard presents the quality of the products over time in a "traffic light system", the quality being represented in three colours:
   - Green = product quality is within expectations
   - Orange = one of the indicators is not within expectations, a quality issue is probable
   - Red = more than one of the indicators are not within expectations, there is a quality issue.
3. **Validation with available in situ data**. Annual validation of the products using the most recent in situ data available. This analysis is necessary in order to retain the validation stage 4 as defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellites (CEOS) (https://lpvs.gsfc.nasa.gov/).
4. **Annual update of the Quality Assessment and Validation Reports**. A summary of the comparison with reference dataset as well as the analysis of quantitative metrics and the results of the validation with in-situ data will be added as an update of the existing quality assessment and validation reports.

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